

CSS3 scroll-snap Properties

Complete Guide to Scroll Snapping and Carousel Effects

CSS Scrolling Mastery

May 10, 2026

Contents

1	Introduction to scroll-snap	2
1.1	What is scroll-snap?	2
1.2	Why Use scroll-snap?	2
1.3	Browser Compatibility	3
2	scroll-snap-type Property	4
2.1	What is scroll-snap-type?	4
2.2	scroll-snap-type Values	5
2.3	Practical scroll-snap-type Examples	5
3	scroll-snap-align Property	6
3.1	What is scroll-snap-align?	6
3.2	Understanding scroll-snap-align	7
3.3	Practical Examples	7
4	scroll-snap-stop Property	8
4.1	What is scroll-snap-stop?	8
4.2	scroll-snap-stop Values	8
4.3	Practical Use Cases	8
5	Practical Carousel Examples	10
5.1	Example 1: Horizontal Image Carousel	10
5.2	Example 2: Vertical Page Scroller	11
5.3	Example 3: Product Gallery Grid	12
6	Advanced scroll-snap Techniques	14
6.1	scroll-padding Property	14
6.2	Combining with scroll-behavior	14
6.3	Responsive Snap Points	15
6.4	Accessibility Considerations	15
7	Real-World Examples	17
7.1	Example: Full-Screen Vertical Sections	17
7.2	Example: Product Showcase Carousel	18
8	Quick Reference	20
8.1	All scroll-snap Properties	20
8.2	Common Patterns	20

8.3	Production Template	21
-----	-------------------------------	----

Chapter 1

Introduction to scroll-snap

1.1 What is scroll-snap?

The CSS scroll-snap specification provides a way to control the scroll behavior of elements by snapping to specific positions. It enables creating responsive carousels, galleries, and paginated content without JavaScript.

Key scroll-snap properties:

- **scroll-snap-type:** Defines container snap behavior
- **scroll-snap-align:** Sets alignment of snap points
- **scroll-snap-stop:** Forces stop at snap point
- **scroll-padding:** Offset for scroll snap positions
- Works on both vertical and horizontal scrolling
- Part of CSS Scroll Snap Module Level 1
- Widely supported in modern browsers
- Enables smooth carousel experiences

1.2 Why Use scroll-snap?

Scroll snapping provides significant UX benefits:

- **Better UX:** Helps users focus on content sections
- **No JavaScript:** Pure CSS implementation
- **Mobile friendly:** Natural touch scrolling experience
- **Accessibility:** Easier navigation for all users
- **Responsive carousels:** Automatic alignment on any screen
- **Performance:** No runtime overhead
- **Progressive enhancement:** Works across browsers

1.3 Browser Compatibility

Scroll-snap is supported in modern browsers:

- Chrome 69+ (2018)
- Firefox 68+ (2019)
- Safari 11+ (2017)
- Edge 79+ (2020)
- Opera 56+ (2018)
- Mobile browsers (most modern versions)

Chapter 2

scroll-snap-type Property

2.1 What is scroll-snap-type?

The `scroll-snap-type` property defines whether an element's scroll container should snap to specific positions.

```
/* Syntax */
scroll-snap-type: [x | y | block | inline | both] [mandatory | proximity];

/* Examples */
/* Vertical mandatory snapping */
.container {
  scroll-snap-type: y mandatory;
}

/* Horizontal mandatory snapping */
.carousel {
  scroll-snap-type: x mandatory;
}

/* Proximity snapping (optional) */
.gallery {
  scroll-snap-type: y proximity;
}

/* Both directions */
.both-snap {
  scroll-snap-type: both mandatory;
}
```

2.2 scroll-snap-type Values

Value	Description
x	Snap horizontally
y	Snap vertically
block	Snap along block axis (vertical)
inline	Snap along inline axis (horizontal)
both	Snap both horizontally and vertically
mandatory	Browser must snap (strict snapping)
proximity	Browser may snap (loose snapping)

2.3 Practical scroll-snap-type Examples

```
<style>
  /* Vertical scrolling carousel - mandatory */
  .vertical-carousel {
    height: 400px;
    overflow-y: scroll;
    scroll-snap-type: y mandatory;
  }

  /* Horizontal scrolling gallery - mandatory */
  .horizontal-gallery {
    overflow-x: scroll;
    scroll-snap-type: x mandatory;
  }

  /* Optional proximity snapping */
  .infinite-scroll {
    overflow-y: scroll;
    scroll-snap-type: y proximity;
  }

  /* Two-dimensional snapping */
  .grid-scroll {
    overflow: scroll;
    scroll-snap-type: both mandatory;
  }
</style>

<div class="vertical-carousel">
  <div class="item">Item 1</div>
  <div class="item">Item 2</div>
</div>
```

Chapter 3

scroll-snap-align Property

3.1 What is scroll-snap-align?

The `scroll-snap-align` property specifies where an element should snap to within the scroll container.

```
/* Syntax */
scroll-snap-align: [start | center | end | none];

/* Examples */
/* Snap to top of element */
.item {
    scroll-snap-align: start;
}

/* Snap to center of element */
.item {
    scroll-snap-align: center;
}

/* Snap to bottom of element */
.item {
    scroll-snap-align: end;
}

/* No snapping for this element */
.item {
    scroll-snap-align: none;
}
```


3.2 Understanding scroll-snap-align

Value	Effect
start	Element's start aligns with container's start
center	Element's center aligns with container's center
end	Element's end aligns with container's end
none	Element is not a snap point

3.3 Practical Examples

```
<style>
  /* Carousel snapping to start */
  .carousel {
    overflow-x: scroll;
    scroll-snap-type: x mandatory;
  }

  .carousel-item {
    scroll-snap-align: start;
    width: 100%;
  }

  /* Gallery snapping to center */
  .gallery {
    overflow-x: scroll;
    scroll-snap-type: x mandatory;
  }

  .gallery-item {
    scroll-snap-align: center;
    width: 300px;
    flex-shrink: 0;
  }

  /* Full-screen pages snapping to start */
  .page-scroller {
    scroll-snap-type: y mandatory;
  }

  .page {
    scroll-snap-align: start;
    height: 100vh;
  }
</style>

<div class="carousel">
  <div class="carousel-item">Slide 1</div>
  <div class="carousel-item">Slide 2</div>
</div>
```

Chapter 4

scroll-snap-stop Property

4.1 What is scroll-snap-stop?

The `scroll-snap-stop` property forces the scroll to stop at snap points during scrolling.

```
/* Syntax */
scroll-snap-stop: [normal | always];

/* Examples */
/* Normal - may skip snap points */
.item {
  scroll-snap-stop: normal;
}

/* Always - must stop at snap point */
.item {
  scroll-snap-stop: always;
}
```

4.2 scroll-snap-stop Values

Value	Effect
normal	Browser may skip snap points if scrolling quickly
always	Must stop at this snap point

4.3 Practical Use Cases

```
/* Always stop at each slide - strict carousel */
.carousel-strict {
  scroll-snap-type: x mandatory;
}

.slide {
  scroll-snap-align: start;
  scroll-snap-stop: always;
  width: 100%;
}
```

```
}

/* Optional stops - softer experience */
.gallery-loose {
  scroll-snap-type: x proximity;
}

.photo {
  scroll-snap-align: center;
  scroll-snap-stop: normal;
  width: 300px;
}

/* Mix strict and flexible */
.mixed {
  scroll-snap-type: x mandatory;
}

.featured {
  scroll-snap-stop: always;
}

.regular {
  scroll-snap-stop: normal;
}
```

Chapter 5

Practical Carousel Examples

5.1 Example 1: Horizontal Image Carousel

```
<style>
.carousel {
  overflow-x: scroll;
  scroll-snap-type: x mandatory;
  scroll-behavior: smooth;
  width: 100%;
  height: 400px;
  display: flex;
  gap: 1rem;
  padding: 1rem;
}

.carousel-item {
  scroll-snap-align: start;
  scroll-snap-stop: always;
  flex-shrink: 0;
  width: 100%;
  height: 100%;
  border-radius: 8px;
  background: #ddd;
}

.carousel-item img {
  width: 100%;
  height: 100%;
  object-fit: cover;
  border-radius: 8px;
}

/* Hide scrollbar */
.carousel::-webkit-scrollbar {
  height: 4px;
}

.carousel::-webkit-scrollbar-track {
  background: #f1f1f1;
}
```

```
    }

    .carousel::-webkit-scrollbar-thumb {
        background: #888;
        border-radius: 2px;
    }
</style>

<div class="carousel">
  <div class="carousel-item">
    
  </div>
  <div class="carousel-item">
    
  </div>
  <div class="carousel-item">
    
  </div>
</div>
```

5.2 Example 2: Vertical Page Scroller

```
<style>
  html {
    scroll-snap-type: y mandatory;
    scroll-behavior: smooth;
  }

  body {
    margin: 0;
  }

  section {
    scroll-snap-align: start;
    scroll-snap-stop: always;
    width: 100%;
    height: 100vh;
    display: flex;
    align-items: center;
    justify-content: center;
    font-size: 2rem;
  }

  section:nth-child(1) {
    background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
  }

  section:nth-child(2) {
    background: linear-gradient(135deg, #f093fb 0%, #f5576c 100%);
  }

  section:nth-child(3) {
```

```
        background: linear-gradient(135deg, #4facfe 0%, #00f2fe 100%);
    }
</style>

<section>Section 1</section>
<section>Section 2</section>
<section>Section 3</section>
```

5.3 Example 3: Product Gallery Grid

```
<style>
.gallery {
    overflow-x: scroll;
    scroll-snap-type: x proximity;
    scroll-behavior: smooth;
    display: flex;
    gap: 1.5rem;
    padding: 1rem;
    width: 100%;
}

.gallery-item {
    scroll-snap-align: center;
    flex-shrink: 0;
    width: 250px;
    border-radius: 8px;
    background: white;
    box-shadow: 0 2px 8px rgba(0, 0, 0, 0.1);
    overflow: hidden;
}

.gallery-item img {
    width: 100%;
    height: 200px;
    object-fit: cover;
}

.gallery-item-info {
    padding: 1rem;
}

.gallery-item h3 {
    margin: 0;
}

.gallery-item p {
    color: #666;
    margin: 0.5rem 0;
}
</style>

<div class="gallery">
```

```
<div class="gallery-item">
  
  <div class="gallery-item-info">
    <h3>Product 1</h3>
    <p>\$99.99</p>
  </div>
</div>
<!-- More items... -->
</div>
```

Chapter 6

Advanced scroll-snap Techniques

6.1 scroll-padding Property

```
/* Add offset to snap positions */
.container {
    scroll-snap-type: y mandatory;
    scroll-padding: 60px 0 0 0; /* Top offset for fixed header */
}

.item {
    scroll-snap-align: start;
}

/* Different padding for different sides */
.carousel {
    scroll-snap-type: x mandatory;
    scroll-padding: 0 20px; /* Left and right padding */
}

/* Using scroll-padding-top for fixed header */
html {
    scroll-snap-type: y mandatory;
    scroll-padding-top: 80px; /* Height of fixed header */
}
```

6.2 Combining with scroll-behavior

```
/* Smooth scrolling with snapping */
html {
    scroll-snap-type: y mandatory;
    scroll-behavior: smooth;
    scroll-padding-top: 60px;
}

section {
    scroll-snap-align: start;
    scroll-snap-stop: always;
}
```



```
    height: 100vh;
}

/* For carousel with smooth snap */
.carousel {
    overflow-x: scroll;
    scroll-snap-type: x mandatory;
    scroll-behavior: smooth;
}

.carousel-item {
    scroll-snap-align: start;
    scroll-snap-stop: always;
}
```

6.3 Responsive Snap Points

```
/* Desktop: snap to center */
.gallery {
    scroll-snap-type: x mandatory;
}

.gallery-item {
    scroll-snap-align: center;
    width: 300px;
}

/* Tablet: snap to start */
@media (max-width: 1024px) {
    .gallery-item {
        scroll-snap-align: start;
        width: 250px;
    }
}

/* Mobile: snap to start, full width */
@media (max-width: 768px) {
    .gallery-item {
        width: 100%;
    }
}
```

6.4 Accessibility Considerations

```
/* Respect reduced motion preference */
@media (prefers-reduced-motion: reduce) {
    html {
        scroll-snap-type: none;
    }
}
```

```
/* Enable snap for users who don't prefer reduced motion */
@media (prefers-reduced-motion: no-preference) {
  html {
    scroll-snap-type: y mandatory;
  }
}

/* Ensure keyboard navigation works */
a:focus {
  outline: 2px solid #0066cc;
  outline-offset: 2px;
}
```

Chapter 7

Real-World Examples

7.1 Example: Full-Screen Vertical Sections

```
<style>
  html {
    scroll-snap-type: y mandatory;
    scroll-behavior: smooth;
  }

  body {
    margin: 0;
  }

  header {
    position: fixed;
    width: 100%;
    top: 0;
    background: white;
    padding: 1rem;
    z-index: 100;
    box-shadow: 0 2px 4px rgba(0, 0, 0, 0.1);
  }

  section {
    scroll-snap-align: start;
    scroll-snap-stop: always;
    height: 100vh;
    display: flex;
    align-items: center;
    justify-content: center;
    margin-top: 60px;
  }

  section:nth-child(odd) {
    background: #f9f9f9;
  }
</style>

<header>
```

```
<nav>Navigation</nav>
</header>

<section id="home">Home Section</section>
<section id="about">About Section</section>
<section id="services">Services Section</section>
```

7.2 Example: Product Showcase Carousel

```
<style>
  .showcase {
    overflow-x: scroll;
    scroll-snap-type: x mandatory;
    scroll-behavior: smooth;
    display: flex;
    gap: 2rem;
    padding: 2rem;
    width: 100%;
  }

  .product {
    scroll-snap-align: center;
    scroll-snap-stop: always;
    flex-shrink: 0;
    width: 400px;
    background: white;
    border-radius: 12px;
    box-shadow: 0 4px 12px rgba(0, 0, 0, 0.1);
    overflow: hidden;
  }

  .product-image {
    width: 100%;
    height: 300px;
    background: #ddd;
    display: flex;
    align-items: center;
    justify-content: center;
  }

  .product-info {
    padding: 1.5rem;
  }

  .product h3 {
    margin: 0;
    font-size: 1.5rem;
  }

  .product p {
    color: #666;
    margin: 0.5rem 0;
  }
```

```
    }

    .product-price {
      font-size: 1.25rem;
      font-weight: bold;
      color: #0066cc;
      margin-top: 1rem;
    }
  }
</style>

<div class="showcase">
  <div class="product">
    <div class="product-image">Image</div>
    <div class="product-info">
      <h3>Product 1</h3>
      <p>Description</p>
      <div class="product-price">\$299.99</div>
    </div>
  </div>
  <!-- More products... -->
</div>
```

Chapter 8

Quick Reference

8.1 All scroll-snap Properties

```
/* Container properties */
scroll-snap-type: [x | y | block | inline | both] [mandatory | proximity];
scroll-padding: top right bottom left;
scroll-padding-top: value;
scroll-padding-right: value;
scroll-padding-bottom: value;
scroll-padding-left: value;

/* Item properties */
scroll-snap-align: [start | center | end | none];
scroll-snap-stop: [normal | always];
```

8.2 Common Patterns

```
/* Pattern 1: Mandatory horizontal carousel */
.carousel {
  overflow-x: scroll;
  scroll-snap-type: x mandatory;
  scroll-behavior: smooth;
}

.item {
  scroll-snap-align: start;
  scroll-snap-stop: always;
  width: 100%;
}

/* Pattern 2: Proximity vertical gallery */
.gallery {
  overflow-y: scroll;
  scroll-snap-type: y proximity;
}

.image {
```

```
    scroll-snap-align: center;
    height: 300px;
}

/* Pattern 3: Full-screen sections */
html {
    scroll-snap-type: y mandatory;
}

section {
    scroll-snap-align: start;
    scroll-snap-stop: always;
    height: 100vh;
}

/* Pattern 4: With fixed header offset */
html {
    scroll-snap-type: y mandatory;
    scroll-padding-top: 60px;
}
```

8.3 Production Template

```
/* Complete scroll-snap setup */

/* Container */
.snap-container {
    overflow-y: scroll;
    scroll-snap-type: y mandatory;
    scroll-behavior: smooth;
    scroll-padding: 80px 0 0 0;
}

/* Items */
.snap-item {
    scroll-snap-align: start;
    scroll-snap-stop: always;
    min-height: 100vh;
}

/* Accessibility */
@media (prefers-reduced-motion: reduce) {
    .snap-container {
        scroll-snap-type: none;
        scroll-behavior: auto;
    }
}

/* Keyboard navigation */
a:focus, button:focus {
    outline: 2px solid #0066cc;
}
```

```
/* Touch-friendly sizing */
@media (max-width: 768px) {
  .snap-item {
    min-height: 100vh;
    scroll-snap-align: start;
  }
}
```